

Free Vibration Analysis of a Rotating, Pre-twisted, Cambered, and Tapered NACA 6412 Airfoil Compressor Blade

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Abstract

A semi-analytical framework is presented for the free vibration and dynamic stability of rotating compressor blades modelled as pre-twisted, tapered, cambered NACA four-digit airfoil shells. The kinematic description follows the degenerated three-dimensional continuum concept: the blade mid-surface is defined by an exact parametric mapping $\mathbf{r}(x, \eta)$ whose first and second derivatives are evaluated analytically, and the through-thickness position vector is generated by offsetting along the exact unit normal. The three-dimensional Jacobian of this mapping is inverted pointwise, so strains are obtained directly from the Cartesian displacement gradient rotated into a local orthonormal surface frame. No Lamé parameters, curvature tensors, or shell-theory approximations of the metric are introduced. A five-field first-order shear deformation kinematic (three Cartesian translations and two director rotations) provides the displacement ansatz. Unlike conventional degenerated-shell finite elements, the field is discretized globally by the Rayleigh–Ritz method using tensor products of scaled Jacobi polynomials $(1+s)P_i^{(0,2)}(s)$ in the span, which enforce the clamped root exactly and are L_2 -orthonormal, and normalised Legendre polynomials in the chord. A singularity-aware quadrature, using the substitution $\eta = t^2$ to remove the $\sqrt{\eta}$ leading-edge singularity of the NACA thickness law and a subdivision at the maximum-camber station where the camber curvature is discontinuous, ensures spectral integration accuracy. Rotation is treated in the blade-fixed frame: the centrifugal body force is applied as a consistent load, the resulting three-dimensional static equilibrium is solved on the reference geometry, and the full Cauchy prestress is used to assemble the geometric stiffness matrix, yielding a prestress-consistent stiffened operator. Coriolis coupling enters through a skew-symmetric gyroscopic matrix and spin-softening through a centrifugal stiffness matrix. The resulting quadratic eigenvalue problem is solved by an energy-scaled Hermitian reformulation that maps the gyroscopic state matrix to a real skew-symmetric form, eliminating the spurious real parts produced by ill-scaled generalized solvers; when the stiffened operator loses positive-definiteness, a scaled general state solve retains the growth rates that identify divergence-type instability. Residual-based a posteriori checks on both the static and dynamic solutions are reported. The method combines the geometric exactness of degenerated three-dimensional continuum mechanics with the mesh-free efficiency of global Ritz functions and is validated against commercial finite element analyses.

Keywords: rotating blade; pre-twist; degenerated shell; first-order shear deformation theory; Rayleigh–Ritz; Jacobi polynomials; centrifugal stiffening; Coriolis effect; gyroscopic eigenvalue problem; dynamic stability; NACA airfoil.

1. Introduction

1.1 Background

Compressor and turbine blades operate in a rotating frame under centrifugal loads that alter their stiffness, couple their bending and torsional motions, and, through Coriolis acceleration, render the governing operator gyroscopic. Accurate prediction of the rotating natural frequencies and of the Campbell diagram is a prerequisite for avoiding resonant engine orders and for high-cycle-fatigue assessment [1–4]. The subject has a long history. Southwell and Gough [5] gave the first estimate of centrifugal stiffening for airscrew blades. Carnegie [6,7] formulated energy expressions for pre-twisted rotating cantilever blading, and Rao [8] consolidated the beam-based literature. Beam models with pre-twist, taper, and Coriolis coupling have since been refined extensively [9–17], and surveys are available [18,19].

Beam theory is inadequate when the blade aspect ratio is low, the chordwise flexibility is significant, or camber and thickness variation across the chord control the modal content. Plate and shell descriptions were therefore introduced. Dokainish and Rawtani [20] analyzed rotating cantilever plates by finite elements; Sreenivasamurthy and Ramamurti [21] examined the Coriolis effect on rotating low-aspect-ratio plates; Ramamurti and Kielb [22] treated twisted rotating plates; Leissa and co-workers [23–26] developed Ritz solutions for cantilevered shallow shells, twisted rotating blades, and three-dimensional twisted parallelepipeds; Yoo and co-workers [27,28] derived modal characteristics of rotating rectangular plates; Hu et al. [29], Kee and Kim [30], Sinha and Turner [31], and Sun et al. [32,33] extended shell models to pre-twisted, cambered, and staggered blades. Reviews of shell vibration [34,35] document the breadth of this literature.

1.2 Gap in the literature

Two limitations recur in the shell-based rotating-blade literature.

First, the majority of analytical and semi-analytical shell formulations rest on classical two-dimensional shell theories (Love–Kirchhoff, Donnell, Sanders, Reissner–Mindlin resultant theories) [36–39]. These theories express the metric through Lamé parameters A_1, A_2 and principal curvatures R_1, R_2 , which are available in closed form only for canonical geometries (cylindrical, conical, spherical, shallow doubly-curved panels). A pre-twisted, tapered, cambered airfoil surface has non-orthogonal parametric lines, non-constant Gaussian curvature, and a thickness that varies in both directions. Fitting such a surface into a Lamé-parameter framework requires either approximating the geometry (shallow-shell or “twisted plate” assumptions) or deriving lengthy problem-specific curvature expressions that are prone to error and do not generalize.

Second, formulations that do treat the geometry exactly, the degenerated-continuum shell elements of Ahmad, Irons and Zienkiewicz [40] and their descendants [41–45] are implemented almost exclusively within the finite element method. The geometric exactness is achieved by interpolating nodal coordinates and nodal directors, which ties the formulation to a mesh, introduces the well-known locking pathologies of low-order shell elements, and forfeits the compactness and convergence rate of global approximation.

The Rayleigh–Ritz method with orthogonal polynomial admissible functions [46–50] delivers exponential convergence for smooth problems with a small number of degrees of freedom and has been applied successfully to plates and shells of canonical geometry. Its use for rotating blades has, however, remained coupled to classical shell kinematics, and its potential in combination with an exact three-dimensional mapping has not been exploited.

1.3 Proposed approach and contributions

This paper closes the gap by combining the two ingredients. The blade is described as a degenerated three-dimensional continuum: the mid-surface position vector and its analytic derivatives, the exact unit normal and its derivatives, and the through-thickness offset define a three-dimensional Jacobian $\mathbf{J}(\mathbf{x}, \eta, \zeta)$ that is inverted at every quadrature point. Strains are then the components of the exact Cartesian displacement gradient, rotated into a local orthonormal frame, with the transverse normal strain removed by the plane-stress condition. The displacement field follows a five-parameter first-order shear deformation theory (FSDT) [36,37,51]: three global Cartesian translations of the mid-surface and two director rotations expressed in the local frame. The unknown fields are expanded globally over the whole blade with L_2 -orthonormal Jacobi–Legendre tensor products; the span functions vanish identically at the root, so the clamped condition is satisfied exactly for all five fields.

Rotation is treated in the blade-fixed frame [52–54]. The centrifugal body force is integrated consistently to a load vector, a three-dimensional static equilibrium problem is solved, the resulting Cauchy stress tensor is evaluated at every volume quadrature point, and the initial-stress (geometric) stiffness matrix is assembled from it. The Coriolis acceleration produces a skew-symmetric gyroscopic matrix and the centripetal acceleration a spin-softening matrix. The quadratic eigenvalue problem is solved by an energy-scaled reformulation that exploits the structure of conservative gyroscopic systems [55–58].

The contributions are:

1. A Lamé-parameter-free shell formulation for pre-twisted, tapered, cambered airfoil blades in which the geometry enters solely through $\mathbf{r}(\mathbf{x}, \eta)$ and its analytic derivatives.
2. A global Rayleigh–Ritz discretization of a degenerated-continuum shell using orthonormal Jacobi and Legendre polynomials, with exact enforcement of the clamped root and a well-conditioned mass matrix by construction.
3. A singularity-aware quadrature for the NACA thickness distribution that recovers spectral integration accuracy despite the $\sqrt{\eta}$ leading-edge behaviour and the camber-curvature discontinuity.
4. A prestress-consistent treatment of centrifugal stiffening in which the full three-dimensional stress state from a static equilibrium solution drives the geometric stiffness.
5. An energy-scaled Hermitian gyroscopic eigensolver that yields exactly imaginary eigenvalues for the stable regime and switches to a scaled general solve that retains growth rates when stiffness loses positive-definiteness, together with residual-based verification of every eigenpair.

The paper is organized as follows. Section 2 defines the blade geometry. Section 3 develops the kinematics, constitutive law, rotational energies, and the prestress-consistent geometric stiffness. Section 4 details the Ritz basis, the quadrature, the matrix assembly, and the eigensolver. Section 5 presents the verification and validation program and the parametric results. Section 6 concludes.

2. Blade Geometry

2.1 Coordinate systems

A right-handed global Cartesian frame (X, Y, Z) with unit vectors $\mathbf{e}_X, \mathbf{e}_Y, \mathbf{e}_Z$ is fixed to the rotor and rotates with it at constant angular velocity $\boldsymbol{\Omega} = \Omega \mathbf{e}_Y$, $\Omega = 2\pi N/60$, where N is the rotor speed in rpm. The blade span is directed along X (radial direction), starting at the hub radius R_h . The stacking (pre-twist) axis is the radial line $Y = Z = 0$.

The mid-surface is parametrized by the span coordinate $x \in [0, L]$ and the dimensionless chord coordinate $\eta \in [0, 1]$, measured from the leading edge ($\eta = 0$) to the trailing edge ($\eta = 1$). The through-thickness coordinate is $\zeta \in \left[-\frac{h}{2}, \frac{h}{2}\right]$.

2.2 Section profile

The section is a NACA four-digit airfoil [59,60] with maximum camber m (first digit, per cent of chord), camber position p (second digit, tenths of chord), and maximum thickness τ (last two digits, per cent of chord). The camber line is

$$f(\eta) = \begin{cases} \frac{m}{p^2} (2p\eta - \eta^2), & \eta \leq p, \\ \frac{m}{(1-p)^2} (1 - 2p + 2p\eta - \eta^2), & \eta > p, \end{cases} \quad f''(\eta) = \begin{cases} -\frac{2m}{p^2}, & \eta \leq p, \\ -\frac{2m}{(1-p)^2}, & \eta > p, \end{cases} \quad (1)$$

so that f and f' are continuous but f'' jumps at $\eta = p$. The thickness distribution is

$$h_0(\eta) = 10 \tau c (0.2969 \sqrt{\eta} - 0.1260 \eta - 0.3516 \eta^2 + 0.2843 \eta^3 + a_4 \eta^4), \quad (2)$$

with $a_4 = -0.1015$ for the standard finite trailing-edge thickness and $a_4 = -0.1036$ for a closed trailing edge. The $\sqrt{\eta}$ term produces the rounded leading edge and an integrable singularity in $\frac{\partial h_0}{\partial \eta}$ at $\eta = 0$.

2.3 Spanwise variation: taper and pre-twist

The chord tapers linearly, $c(x) = c_0 \left(1 - \tau_c \frac{x}{L}\right)$, and the thickness is scaled independently, $h(x, \eta) = h_0(x, \eta) \left(1 - \tau_h \frac{x}{L}\right)$, where c_0 is the root chord and $\tau_c, \tau_h \in [0, 1)$ are the chord and thickness taper fractions. The section is rotated about the stacking axis by the linearly varying angle

$$\theta(x) = \theta_0 + \kappa x, \quad \kappa = \frac{\theta_{\text{tip}}}{L}, \quad (3)$$

where θ_0 is the stagger angle at the root and θ_{tip} the total pre-twist. The section-fixed unit vectors are

$$\mathbf{e}_y(x) = \begin{bmatrix} 0 \\ \cos \theta(x) \\ \sin \theta(x) \end{bmatrix}, \quad \mathbf{e}_z(x) = \begin{bmatrix} 0 \\ -\sin \theta(x) \\ \cos \theta(x) \end{bmatrix}, \quad (44)$$

The stacking axis pierces each section on its camber line at the chord fraction η_a ; the camber line is shifted so that this point coincides with the origin of the section plane, $\hat{f}(\eta) = f(\eta) - f(\eta_a)$, and the chordwise abscissa is $a(\eta) = \eta - \eta_a$.

2.4 Mid-surface mapping and analytic derivatives

The mid-surface position vector is

$$\mathbf{r}(x, \eta) = (R_h + x) \mathbf{e}_x + c(x) [a(\eta) \mathbf{e}_y(x) + \hat{f}(\eta) \mathbf{e}_z(x)]. \quad (5)$$

Introducing the auxiliary vectors $\mathbf{v} = a \mathbf{e}_y + \hat{f} \mathbf{e}_z$ and $\mathbf{w} = a \mathbf{e}_z - \hat{f} \mathbf{e}_y$ (so that $\frac{\partial \mathbf{v}}{\partial x} = \kappa \mathbf{w}$), the first derivatives are

$$\mathbf{r}_{,x} = \mathbf{e}_x + c' \mathbf{v} + \kappa c \mathbf{w}, \quad \mathbf{r}_{,\eta} = c(\mathbf{e}_y + f' \mathbf{e}_z), \quad (6)$$

with $c' = -\frac{c_0 \tau_c}{L}$, and the second derivatives are

$$\mathbf{r}_{,xx} = 2c' \kappa \mathbf{w} - c\kappa^2 \mathbf{v}, \quad \mathbf{r}_{,x\eta} = c'(e_y + f'e_z) + c\kappa(\mathbf{e}_z - f'e_y), \quad \mathbf{r}_{,\eta\eta} = cf'' \mathbf{e}_z. \quad (75)$$

The unit normal and the surface Jacobian are

$$\mathbf{n} = \frac{\mathbf{r}_{,x} \times \mathbf{r}_{,\eta}}{j}, \quad j = \|\mathbf{r}_{,x} \times \mathbf{r}_{,\eta}\| > 0, \quad (86)$$

and the derivatives of the normal follow from differentiating Eq. (8) and projecting out the component along $\mathbf{n}$:

$$\mathbf{n}_{,\alpha} = \frac{1}{j} [\mathbf{c}_{,\alpha} - \mathbf{n}(\mathbf{n} \cdot \mathbf{c}_{,\alpha})], \quad \mathbf{c}_{,x} = \mathbf{r}_{,xx} \times \mathbf{r}_{,\eta} + \mathbf{r}_{,x} \times \mathbf{r}_{,x\eta}, \quad \mathbf{c}_{,\eta} = \mathbf{r}_{,x\eta} \times \mathbf{r}_{,\eta} + \mathbf{r}_{,x} \times \mathbf{r}_{,\eta\eta}. \quad (97)$$

2.5 Local orthonormal frame

At every mid-surface point a right-handed orthonormal triad $\mathbf{Q} = [\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3]$ is constructed with

$$\mathbf{e}_1 = \frac{\mathbf{r}_{,x}}{\|\mathbf{r}_{,x}\|}, \quad \mathbf{e}_3 = \mathbf{n}, \quad \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_1. \quad (108)$$

Its derivatives $\mathbf{Q}_{,x}, \mathbf{Q}_{,\eta}$ are obtained analytically by the same projection rule applied to $\mathbf{r}_{,xx}, \mathbf{r}_{,x\eta}$ and by the product rule for the cross product in Eq. (10). The availability of $\mathbf{Q}_{,\alpha}$ in closed form is essential: the director rotations of the FSDT kinematic are resolved in this frame, and their spatial derivatives require the derivatives of the frame itself. No Lamé parameters, Christoffel symbols, or curvature tensors appear anywhere in the formulation; the geometry is fully specified by $\mathbf{r}, \mathbf{r}_{,\alpha}, \mathbf{r}_{,\alpha\beta}$.

3. Mathematical Formulation

3.1 Degenerated-continuum kinematics

The position of a material point in the reference configuration is

$$\mathbf{X}(x, \eta, \zeta) = \mathbf{r}(x, \eta) + \zeta \mathbf{n}(x, \eta), \quad (119)$$

and the three-dimensional Jacobian of the mapping $(x, \eta, \zeta) \mapsto \mathbf{X}$ is

$$\mathbf{J} = [\mathbf{r}_{,x} + \zeta \mathbf{n}_{,x} \mid \mathbf{r}_{,\eta} + \zeta \mathbf{n}_{,\eta} \mid \mathbf{n}], \quad \det \mathbf{J} > 0. \quad (1210)$$

The determinant is monitored at every quadrature point; a non-positive value indicates that the thickness exceeds the local radius of curvature and the shell description is invalid.

The displacement field follows a five-parameter FSDT [36,37,51]:

$$\mathbf{u}(x, \eta, \zeta) = \mathbf{u}_0(x, \eta) + \zeta[\varphi_1(x, \eta) \mathbf{e}_1(x, \eta) + \varphi_2(x, \eta) \mathbf{e}_2(x, \eta)], \quad (1311)$$

where $\mathbf{u}_0 = (u, v, w)$ are the global Cartesian translations of the mid-surface and φ_1, φ_2 are the director rotations, i.e., the components of the director increment in the local tangent plane. The drilling rotation is absent, and the director is inextensible (no thickness stretch), consistent with the plane-stress assumption adopted below.

The partial derivatives with respect to the parametric coordinates are

$$\mathbf{u}_{,x} = \mathbf{u}_{0,x} + \zeta(\varphi_{1,x}\mathbf{e}_1 + \varphi_1\mathbf{e}_{1,x} + \varphi_{2,x}\mathbf{e}_2 + \varphi_2\mathbf{e}_{2,x}), \quad \mathbf{u}_{,\zeta} = \varphi_1\mathbf{e}_1 + \varphi_2\mathbf{e}_2, \quad (1412)$$

with $\mathbf{u}_{,\eta}$ analogous. The terms $\varphi_\alpha\mathbf{e}_{\alpha,x}$ carry the coupling between the director rotations and the twist, taper, and camber of the surface; they are exactly the terms that classical theories approximate through curvature terms.

The Cartesian displacement gradient is $\nabla_X \mathbf{u} = [\mathbf{u}_{,x} \ \mathbf{u}_{,\eta} \ \mathbf{u}_{,\zeta}] \mathbf{J}^{-1}$, and its representation in the local frame is

$$\mathbf{H} = \mathbf{Q}^\top (\nabla_X \mathbf{u}) \mathbf{Q}. \quad (1513)$$

In the implementation, $\mathbf{J}^{-1} \mathbf{Q}$ is formed by one 3×3 linear solve per quadrature point, after which the gradient along the local material directions is available for both the strain operator and the geometric stiffness.

The engineering strain vector used with the plane-stress constitutive law is

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \\ \gamma_{13} \\ \gamma_{23} \end{bmatrix} = \begin{bmatrix} H_{11} \\ H_{22} \\ H_{12} + H_{21} \\ H_{13} + H_{31} \\ H_{23} + H_{32} \end{bmatrix} = \mathbf{B}(x, \eta, \zeta) \mathbf{q}, \quad (1614)$$

where $\mathbf{q}$ collects the Ritz coefficients (Section 4) and $\mathbf{B}$ is the strain–displacement operator. The transverse normal strain ε_{33} is eliminated by the plane-stress condition $\sigma_{33} = 0$. Because ζ is treated as an independent coordinate in the differentiation of Eq. (13), the spanwise and chordwise variation of the thickness enters exclusively through the integration limits and the Jacobian, not through the strain operator, a distinction that is automatic in the degenerated-continuum approach but must be imposed by hand in resultant-based shell theories.

3.2 Constitutive relation

For an isotropic, homogeneous material with Young’s modulus E , Poisson’s ratio ν , and shear modulus $G = \frac{E}{[2(1+\nu)]}$, the plane-stress constitutive matrix in the local frame is

$$\mathbf{C} = \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0 & 0 & 0 \\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 & 0 & 0 \\ 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & k_s G & 0 \\ 0 & 0 & 0 & 0 & k_s G \end{bmatrix}, \quad (17)$$

with the shear correction factor $k_s = \frac{\pi^2}{12}$ [37,61]. The transverse shear correction compensates for the constant through-thickness distribution of γ_{13}, γ_{23} implied by Eq. (13).

3.3 Energy functionals in the rotating frame

Let $\mathbf{S}$ denote the skew-symmetric matrix of $\boldsymbol{\Omega}$, $\mathbf{S}\mathbf{a} = \boldsymbol{\Omega} \times \mathbf{a}$, so that for $\boldsymbol{\Omega} = \Omega \mathbf{e}_Y$,

$$\mathbf{S} = \begin{bmatrix} 0 & 0 & \Omega \\ 0 & 0 & 0 \\ -\Omega & 0 & 0 \end{bmatrix}, \quad \mathbf{S}^\top \mathbf{S} = -\mathbf{S}\mathbf{S}. \quad (18)$$

The absolute acceleration of a material point observed in the rotating frame is

$$\mathbf{a} = \ddot{\mathbf{u}} + 2\mathbf{S}\dot{\mathbf{u}} + \mathbf{S}\mathbf{S}(\mathbf{X} + \mathbf{u}), \quad (19)$$

comprising the relative, Coriolis, and centripetal terms. Writing the discretized displacement as $\mathbf{u} = \mathbf{U}(\mathbf{x}, \eta, \zeta) \mathbf{q}(t)$, where $\mathbf{U}$ is the $3 \times n$ displacement operator built from the Ritz basis and the director frame, the principle of virtual work,

$$\int_V \rho \delta \mathbf{u}^\top \mathbf{a} dV + \int_V \delta \boldsymbol{\varepsilon}^\top \boldsymbol{\sigma} dV = 0, \quad (20)$$

yields the semi-discrete equations of motion

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{G}\dot{\mathbf{q}} + (\mathbf{K}_{\text{el}} + \mathbf{K}_\sigma - \mathbf{K}_\Omega)\mathbf{q} = \mathbf{f}_c, \quad (21)$$

with the elastic stiffness, mass, gyroscopic, spin-softening, and centrifugal load defined by the volume integrals

$$\begin{aligned} \mathbf{K}_{\text{el}} &= \int_V \mathbf{B}^\top \mathbf{C} \mathbf{B} dV, & \mathbf{M} &= \int_V \rho \mathbf{U}^\top \mathbf{U} dV, \\ \mathbf{G} &= 2 \int_V \rho \mathbf{U}^\top \mathbf{S} \mathbf{U} dV, & \mathbf{K}_\Omega &= \int_V \rho \mathbf{U}^\top \mathbf{S}^\top \mathbf{S} \mathbf{U} dV, \\ \mathbf{f}_c &= \int_V \rho \mathbf{U}^\top \mathbf{S}^\top \mathbf{S} \mathbf{X} dV, & dV &= \det \mathbf{J} \cdot \frac{h}{2} d\hat{\zeta} d\eta dx, \quad \hat{\zeta} \in [-1, 1]. \\ \mathbf{K}_\sigma &= \int_0^L \int_0^1 \int_{-1}^1 \sum_{k=1}^3 (\nabla_1 \mathbf{U}_k)^\top \boldsymbol{\Sigma}_0 (\nabla_1 \mathbf{U}_k) \det \mathbf{J} \frac{h}{2} d\hat{\zeta} d\eta dx \end{aligned} \quad (22)$$

$\mathbf{M}$, $\mathbf{K}_{\text{el}}$, $\mathbf{K}_\Omega$ are symmetric and $\mathbf{G}$ is skew-symmetric; these properties are enforced exactly after assembly to remove round-off asymmetry. $\mathbf{K}_\sigma$ is the geometric stiffness derived in Section 3.4. The sign of $\mathbf{K}_\Omega$ reflects spin softening: the centripetal term $\mathbf{S}\mathbf{S}\mathbf{u} = -\mathbf{S}^\top \mathbf{S}\mathbf{u}$ reduces the restoring force for displacements perpendicular to the rotation axis. Because the mass operator $\mathbf{U}$ includes the ζ -weighted director terms, both the translational and rotary inertia of the shell are captured consistently in $\mathbf{M}$, $\mathbf{G}$, and $\mathbf{K}_\Omega$, and the centrifugal load $\mathbf{f}_c$ includes the moment produced by the through-thickness offset of the body force.

3.4 Prestress-consistent geometric stiffness

Centrifugal stiffening arises from the interaction of the steady stress field with the second-order (rotation) part of the strain. Rather than adopting a beam-type axial-force approximation or a membrane-resultant approximation, the present formulation evaluates the complete three-dimensional stress state produced by the centrifugal load and integrates it against the full displacement gradient.

Static equilibrium. The steady deformation $\mathbf{q}_0$ on the reference geometry is obtained from

$$\mathbf{K}_{\text{el}} \mathbf{q}_0 = \mathbf{f}_c, \quad (23)$$

solved in mass-normalized coordinates (Section 4.4) after verification that the balanced stiffness admits a Cholesky factorization; failure indicates that the reference configuration is not a stable static state and that

a nonlinear equilibrium path would be required. The relative residual $\frac{\|\mathbf{K}_{el} \mathbf{q}_0 - \mathbf{f}_c\|}{\|\mathbf{f}_c\|}$ is recorded as a solution diagnostic.

Stress recovery. At every volume quadrature point the strain $\boldsymbol{\varepsilon}_0 = \mathbf{B}\mathbf{q}_0$ and the stress $\boldsymbol{\sigma}_0 = \mathbf{C}\boldsymbol{\varepsilon}_0$ are evaluated and assembled into the symmetric local stress tensor

$$\boldsymbol{\Sigma}_0 = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & 0 \end{bmatrix}. \quad (24)$$

Initial-stress stiffness. Linearizing the Green–Lagrange strain about the prestressed state and retaining the quadratic term yields the classical initial-stress operator [41,45,62]

$$\delta W_\sigma = \int_V \sigma_{ab}^0 \delta u_{k,a} u_{k,b} dV \quad \Rightarrow \quad \mathbf{K}_\sigma = \int_V \sum_{k=1}^3 (\nabla_\ell U_k)^\top \boldsymbol{\Sigma}_0 (\nabla_\ell U_k) dV, \quad (25)$$

where $\nabla_\ell U_k$ is the row of the operator giving the derivative of the k-th Cartesian displacement component along the three local directions. All nine gradient components and all five independent stress components participate; no resultant or beam approximation is introduced. This is the sense in which the formulation is prestress-consistent: the same kinematic operator, the same quadrature, and the same three-dimensional Jacobian are used for the static solution, the stress recovery, and the geometric stiffness.

Validity of the linear reference-state approximation. Equation (23) is a linear solve on the undeformed geometry. Its adequacy is checked a posteriori by three quantities reported for every analysis: the maximum linear strain $\max|\boldsymbol{\varepsilon}_0|$ (flagged above 1 %), the maximum static mid-surface displacement normalised by span (flagged above 2 %), and the maximum director slope $\sqrt{\varphi_1^2 + \varphi_2^2}$ (flagged above 0.05 rad). Exceedance of any threshold indicates that a geometrically nonlinear equilibrium and tangent operator would be required.

3.5 Eigenvalue problem

Setting $\mathbf{f}_c$ aside (it determines $\mathbf{q}_0$ only) and substituting $\mathbf{q} = \boldsymbol{\Phi} \mathbf{e}^{\lambda t}$ into the homogeneous form of Eq. (21) gives the quadratic eigenvalue problem

$$(\lambda^2 \mathbf{M} + \lambda \mathbf{G} + \mathbf{K}) \boldsymbol{\Phi} = \mathbf{0}, \quad \mathbf{K} = \mathbf{K}_{el} + \mathbf{K}_\sigma - \mathbf{K}_\Omega. \quad (26)$$

The natural frequency of a mode is $f = \frac{\text{Im}(\lambda)}{2\pi}$ and its growth rate is $\text{Re}(\lambda)$. For $\mathbf{K} \succ 0$ the system is a conservative gyroscopic system whose eigenvalues are purely imaginary and whose motion is stable [55,56,63,64]. Loss of positive-definiteness of $\mathbf{K}$ -which spin softening can produce at sufficiently high speed-admits eigenvalues with positive real part, corresponding to divergence-type dynamic instability. Detecting and quantifying this transition is one objective of the present solver.

4. Computational Implementation

4.1 Rayleigh–Ritz basis

Each of the five fields $\{u, v, w, \varphi_1, \varphi_2\}$ is expanded in the same tensor-product basis of $n_s \times n_c$ functions,

$$\psi_{ij}(x, \eta) = X_i(x) Y_j(\eta), \quad i = 0, \dots, n_s - 1, \quad j = 0, \dots, n_c - 1, \quad (27)$$

giving $n = 5 n_s n_c$ degrees of freedom.

Span functions. With $s = \frac{2x}{L} - 1 \in [-1, 1]$,

$$X_i(x) = \sqrt{\frac{2i+3}{8}} (1+s) P_i^{(0,2)}(s), \quad (28)$$

where $P_i^{(\alpha,\beta)}$ is the Jacobi polynomial [65,66]. The factor $(1+s)$ vanishes at the root $x = 0$, so every field—translations and director rotations alike—satisfies the fully clamped condition identically, with no penalty or constraint equation. The Jacobi weight $(1+s)^\beta$ with $\beta = 2$ absorbs the square of this factor, so that

$$\int_{-1}^1 X_i X_k ds = \delta_{ik}, \quad (29)$$

by the Jacobi orthogonality relation $\int (1+s)^2 P_i^{(0,2)} P_k^{(0,2)}(s) ds = \frac{8}{(2i+3)} \delta_{ik}$. The derivative is evaluated analytically through $\frac{d}{ds} P_i^{(0,2)}(s) = \frac{1}{2}(i+3) P_{i-1}^{(1,3)}(s)$. The basis is thus a set of L_2 -orthonormal admissible functions in the sense of Bhat [46] and Liew and Lam [47], obtained here in closed form rather than by Gram–Schmidt recursion, which avoids the loss of orthogonality that accumulates in numerically generated sequences [50].

Chord functions. With $t = 2\eta - 1$,

$$Y_j(\eta) = \sqrt{\frac{2j+1}{2}} P_j(t), \quad (30)$$

the normalized Legendre polynomials, with $\frac{d}{dt} P_j = \frac{1}{2}(j+1) P_{j-1}^{(1,1)}$. The leading and trailing edges are free; the natural boundary conditions are satisfied variationally.

The orthonormality of both families produces a mass matrix that, for a plate of uniform thickness and orthogonal parametric lines, would be exactly diagonal; for the actual twisted, tapered airfoil it remains strongly diagonally dominant. This is the structural origin of the well-conditioned mass-normalization step in Section 4.4 and is a decisive advantage over monomial or trigonometric Ritz bases, whose Gram matrices become numerically singular at moderate order.

Displacement operator. For the translational fields the operator is $\mathbf{U}^{(k)} = \mathbf{e}_k \otimes \boldsymbol{\Psi}$ with constant global direction $\mathbf{e}_k$; for the rotational fields it is $\mathbf{U}^{(3+\alpha)} = \zeta \mathbf{e}_\alpha(x, \eta) \otimes \boldsymbol{\Psi}$, whose parametric derivatives require $\mathbf{e}_{\alpha,x}, \mathbf{e}_{\alpha,\eta}$ from Section 2.5. The derivatives in Eq. (14) are taken at fixed η , consistent with the surface parametrisation.

4.2 Singularity-aware quadrature

The integrands in Eqs. (22) and (25) are smooth in x , and $n_q^{(x)}$ Gauss–Legendre points on $[0, L]$ suffice. In the chord direction two features degrade the convergence of naive Gauss quadrature: the $\sqrt{\eta}$ behaviour of the thickness, Eq. (2), which enters the volume element through $h/2$ and the mass and stiffness through h and h^3 -type moments, and the jump in f'' at $\eta = p$, which enters through $\mathbf{r}_{,\eta\eta}$, hence $\mathbf{n}_{,\eta}$, the frame derivatives, and $\det \mathbf{J}$.

Leading-edge regularization. The substitution $\eta = t^2$, $d\eta = 2t dt$, maps $\sqrt{\eta} \rightarrow t$ and renders every integrand a smooth function of t on $[0,1]$; the Jacobian $2t$ of the substitution is absorbed into the quadrature weights. Gauss–Legendre quadrature in t then converges exponentially.

Camber-curvature split. The interval in t is divided at $t = \sqrt{p}$ so that each sub-interval contains a polynomially smooth camber line. $n_q^{(\eta)}$ Gauss points are placed on each sub-interval; for a symmetric section ($m = 0$) no split is applied.

Thickness integration. $n_q^{(\zeta)}$ Gauss–Legendre points are used on $\hat{\zeta} \in [-1,1]$, with $\zeta = \hat{\zeta} h(x, \eta)/2$. The integrands in ζ are polynomial of degree at most two for a straight director (the ζ^2 rotary-inertia and bending terms) multiplied by the rational function $\det \mathbf{J}(\zeta)$; three points are sufficient for the thin-to-moderately-thick regime, and the requirement $n_q^{(\zeta)} \geq 2$ is enforced so that rotary inertia is captured.

The total number of volume quadrature points is $n_q^{(x)} n_q^{(\eta)} n_{\text{sub}} n_q^{(\zeta)}$, where $n_{\text{sub}} = 2$ for a cambered section and 1 otherwise. The requirement $n_q^{(x)} \geq n_s + 1$, $n_q^{(\eta)} \geq n_c + 1$ is enforced to avoid rank deficiency of the discrete operators.

4.3 Matrix assembly

All operators are assembled in a single pass over the volume quadrature points, organized by thickness layer to bound peak memory. At each layer the displacement operator $\mathbf{U}$, the local gradient operator, the strain operator $\mathbf{B}$, $\det \mathbf{J}$, and the material position $\mathbf{X}$ are evaluated in vectorised form for all surface points, and the contributions

$$\mathbf{B}^T \mathbf{C} \mathbf{B}, \quad \rho \mathbf{U}^T \mathbf{U}, \quad 2\rho \mathbf{U}^T \mathbf{S} \mathbf{U}, \quad \rho (\mathbf{S} \mathbf{U})^T (\mathbf{S} \mathbf{U}), \quad \rho \mathbf{U}^T \mathbf{S}^T \mathbf{S} \mathbf{X}$$

are accumulated with the weights $w = w_x w_\eta w_\zeta (h/2) \det \mathbf{J}$. The blade volume $\int dV$ and mass $\rho \int dV$ are accumulated simultaneously and compared with the CAD volume as a first-level check on the mapping and quadrature. After the static solution, a second pass over the same points evaluates Σ_0 and assembles $\mathbf{K}_\sigma$ by Eq. (25).

4.4 Mass normalization

The generalized problem is reduced to standard form through a congruence transformation $\mathbf{q} = \mathbf{T} \mathbf{y}$ with $\mathbf{T}^T \mathbf{M} \mathbf{T} = \mathbf{I}$. To preserve accuracy across the disparate scales of translational and ζ -weighted rotational degrees of freedom, $\mathbf{M}$ is first diagonally equilibrated, $\tilde{\mathbf{M}} = \mathbf{D} \mathbf{M} \mathbf{D}$ with $\mathbf{D} = \text{diag}(\mathbf{M}_{ii}^{-1/2})$, then factorised, $\tilde{\mathbf{M}} = \mathbf{L} \mathbf{L}^T$, giving $\mathbf{T} = \mathbf{D} \mathbf{L}^{-T}$ [67]. The same transformation is used for the static solve, Eq. (23), so that the balanced stiffness $\mathbf{T}^T \mathbf{K}_{el} \mathbf{T}$ can be tested for positive-definiteness by Cholesky factorisation before the solution is accepted.

4.5 Energy-scaled Hermitian gyroscopic eigensolver

In mass-normalized coordinates the problem is $(\lambda^2 \mathbf{I} + \lambda \tilde{\mathbf{G}} + \tilde{\mathbf{K}}) \mathbf{y} = \mathbf{0}$ with $\tilde{\mathbf{K}}$ symmetric and $\tilde{\mathbf{G}}$ skew-symmetric. Its spectral decomposition $\tilde{\mathbf{K}} = \mathbf{V} \mathbf{A} \mathbf{V}^T$ is computed by a symmetric solver.

Positive-definite stiffness. If $\lambda_{\min}(\tilde{\mathbf{K}}) > 0$, the state vector is scaled by the square-root stiffness, $\mathbf{z} = \{\Lambda^{1/2} \mathbf{V}^T \mathbf{y}; \mathbf{V}^T \dot{\mathbf{y}}\}$, so that both partitions carry the dimension of the square root of energy. The first-order system becomes

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{\Lambda}^{1/2} \\ -\mathbf{\Lambda}^{1/2} & -\mathbf{V}^T \tilde{\mathbf{G}} \mathbf{V} \end{bmatrix}, \quad (31)$$

which is exactly real skew-symmetric. Consequently $i\mathbf{A}$ is Hermitian, its eigenvalues μ are real, and the eigenvalues of Eq. (26) are $\lambda = -i\mu$ —purely imaginary to machine precision. This reformulation, which follows from the energy structure of conservative gyroscopic systems [55–57], has three consequences that a direct application of a generalized (QZ) solver to the unscaled $2n \times 2n$ state matrix does not share: (i) the eigenvalues carry no spurious real parts, so stability classification is unambiguous; (ii) repeated frequencies are returned with complete eigenspaces, since Hermitian solvers are backward stable for clustered eigenvalues; (iii) the cost is that of one Hermitian eigensolve of order $2n$, with eigenvectors orthonormal in the energy inner product. The modal vectors are recovered by $\boldsymbol{\Phi} = \mathbf{T}\mathbf{V}\mathbf{\Lambda}^{-1/2}\mathbf{z}_{(1)}$ where $\mathbf{z}_{(1)}$ is the upper partition.

Indefinite stiffness. If $\lambda_{\min}(\tilde{\mathbf{K}}) \leq 0$, the square-root scaling is unavailable and instability is possible. The solver then forms the conventional companion matrix with a time scaling $\tau = \max(\sqrt{\max|\Lambda_i|}, \|\tilde{\mathbf{G}}\|_\infty, 1)$,

$$\mathbf{A}_\tau = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\frac{\tilde{\mathbf{K}}}{\tau^2} & -\frac{\tilde{\mathbf{G}}}{\tau} \end{bmatrix}, \quad \lambda = \tau \mu(\mathbf{A}_\tau), \quad (32)$$

and applies a general (non-symmetric) eigensolver [67,68]. Eigenvalues with $\text{Re}\lambda > 10^{-7} + 10^{-8}|\lambda|$ are classified as growing modes. The method used—Hermitian or general—is recorded with each result, together with $\lambda_{\min}(\tilde{\mathbf{K}})$ as the margin to instability.

A posteriori verification. Every eigenpair is normalized to unit modal mass, its complex phase is fixed by the largest component for reproducible reporting, and its relative residual

$$r = \frac{\|(\lambda^2 \mathbf{M} + \lambda \mathbf{G} + \mathbf{K})\boldsymbol{\Phi}\|}{\|\mathbf{K}\boldsymbol{\Phi}\| + |\lambda| \|\mathbf{G}\boldsymbol{\Phi}\| + |\lambda|^2 \|\mathbf{M}\boldsymbol{\Phi}\|} \quad (33)$$

is evaluated on the original (untransformed) matrices [69]. Modes are reported in ascending order of $\text{Im}\lambda$; residuals exceeding 10^{-6} are flagged.

4.6 Software and reproducibility

The formulation is implemented in Python using NumPy [70] and SciPy [71] (LAPACK-backed dense linear algebra [68], Jacobi and Legendre evaluation) with Matplotlib [72] for post-processing. For each run the code exports the frequency table with growth rates and residuals, the complete complex spectrum, the assembled operators, the static coefficient vector, and a JSON record of the configuration and all diagnostics. Root and tip section point clouds are exported in a CAD-compatible tabular format to permit construction of a geometrically identical solid model for finite element validation.

4.7 Reference configuration

Unless stated otherwise, results refer to the reference blade defined in Table 1.

Table 1. Reference blade configuration.

Quantity	Symbol	Value
Span	L	0.3 m
Root chord	c_0	0.2 m

Quantity	Symbol	Value
Section	—	NACA 6412 (finite trailing edge)
Chord taper fraction	τ_c	0.10
Thickness taper fraction	τ_h	0.50
Root stagger	θ_0	-20°
Total pre-twist	θ_{tip}	-15°
Stacking axis chord fraction	η_a	0.25
Hub radius	R_h	0.2 m
Rotor speed	N	4000 rpm
Young's modulus	E	200 GPa
Poisson's ratio	ν	0.30
Density	ρ	7850 kg m^{-3}
Shear correction factor	k_s	$\pi^2/12$
Ritz order	$n_s \times n_c$	6×6 (150 DOF)
Quadrature	$n_q^{(x)}, n_q^{(\eta)}, n_q^{(\zeta)}$	10, 10 (per sub-interval), 3

5. Results and Discussion

5.1 Geometric verification

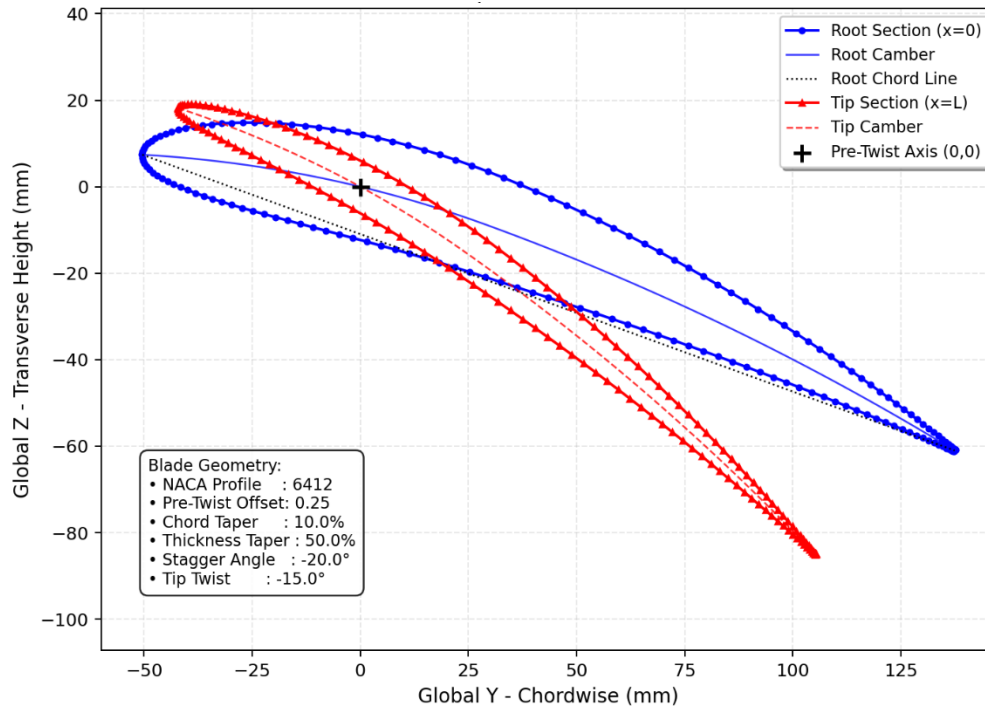


Figure 1 - Root and tip section geometry: Overlay of the analytically generated root ($x = 0$) and tip ($x = L$) sections of the reference blade in the section plane, showing upper and lower skins, camber lines, chord lines, and the stacking axis at the origin. The tip section exhibits the 10 % chord reduction, 50 %

thickness reduction, and -15° relative rotation about the stacking axis. Superimpose the corresponding sections extracted from the CAD model used for finite element validation and report the maximum point-wise deviation in mm.

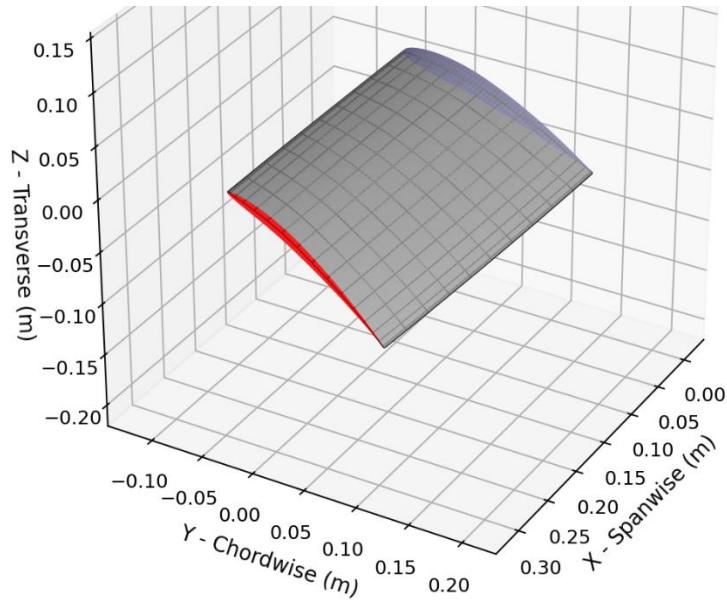


Figure 2 - Three-dimensional undeformed blade: Isometric rendering of the mid-surface and skins generated by Eq. (11) with the hub-radius offset, illustrating the combined pre-twist, taper, and camber. Annotate the rotation axis $\mathbf{e}_Y$ and the span direction.

Table 2 - Volume and mass check: Blade volume and mass computed by the singularity-aware quadrature (Section 4.2) versus the CAD solid model, for the reference configuration and for a symmetric NACA 0012 variant. Report the relative discrepancy and demonstrate that it is below the quadrature tolerance. Include a column showing the volume error obtained without the $\eta = t^2$ substitution at the same quadrature order, to quantify the benefit of the leading-edge regularization.

Parameter	ANSYS	Code-RR	% Difference	Parameter
Mass	5.36798535	5.35611030	-0.2212%	Mass
Volume	0.00068382	0.00068231	-0.2208%	Volume

5.2 Convergence

Table - Ritz convergence of the lowest Seven frequencies

Modes	2	3	4	5	6	7	8	10
1	551.47	267.97	248.87	246.86	246.01	245.66	245.57	245.41
2	1671.58	788.12	724.04	712.70	708.17	706.49	705.83	704.90
3	2409.66	1180.70	1005.19	972.58	969.46	968.41	968.15	967.91
4	4124.04	1659.64	1623.39	1609.34	1606.47	1605.21	1604.81	1604.48
5	4967.57	2101.84	1734.83	1664.66	1657.86	1655.06	1654.52	1653.75
6	5139.46	5012.99	2464.40	2226.00	2116.92	2103.02	2094.54	2092.70
7	5958.40	5230.24	2906.82	2446.49	2413.46	2405.94	2404.19	2403.06

Figure 3 - Ritz convergence of the lowest six non-rotating frequencies: Frequencies versus polynomial order $n_s = n_c \in \{3, \dots, 10\}$ at $N = 0$, plotted on a semi-logarithmic scale as relative change between successive orders. Demonstrate the exponential convergence characteristic of global orthogonal bases and identify the order at which all six frequencies change by less than 0.1 %. Report the condition number of $\mathbf{M}$ before and after diagonal equilibration at each order to illustrate the well-posedness of the orthonormal basis.

[PLACEHOLDER: Figure 4 — Quadrature convergence.] Caption: Lowest six frequencies versus chordwise quadrature order $n_q^{(\eta)}$ at fixed Ritz order, (a) with the $\eta = t^2$ substitution and camber split, (b) with plain Gauss–Legendre quadrature on η . Show the algebraic convergence of (b), attributable to the $\sqrt{\eta}$ singularity, against the rapid saturation of (a). Include a third curve showing convergence with respect to $n_q^{(\zeta)} \in \{2, 3, 4, 5\}$ to justify the three-point thickness rule.

[PLACEHOLDER: Table 3 — Eigenpair residuals.] Caption: Maximum relative eigenpair residual, Eq. (33), across all modes at the converged discretization for $N \in \{0, 2000, 4000, 6000\}$ rpm, together with the solver branch used (Hermitian or general) and $\lambda_{\min}(\tilde{\mathbf{K}})$. All residuals should be well below the 10^{-6} acceptance threshold.

5.3 Validation against finite element analysis: non-rotating blade

[PLACEHOLDER: Table 4 — Non-rotating frequency comparison with ANSYS.] Caption: Lowest eight natural frequencies (Hz) of the reference blade at $N = 0$: present method (150 DOF) versus ANSYS Mechanical using (a) quadratic shell elements (SHELL281) on the mid-surface with the exact thickness distribution applied as a field, and (b) quadratic solid elements (SOLID186) on the CAD solid. State the element counts and degrees of freedom of each FE model, the mesh-convergence tolerance achieved, and the percentage difference of the present method from each. Discuss the residual differences in terms of the plane-stress assumption, the shear correction factor, and the finite trailing-edge face modelled in the solid.

[PLACEHOLDER: Figure 5 — Non-rotating mode shapes: present versus ANSYS.] Caption: Side-by-side deformed skins for the lowest six modes, colored by displacement magnitude on a common scale. Identify each mode by its dominant character (first flap, first edgewise, first torsion, second flap, chordwise bending, coupled flap–torsion) and comment on the flap–torsion coupling induced by pre-twist and the stacking-axis offset from the section centroid. Report the Modal Assurance Criterion between corresponding present and FE mode shapes evaluated on a common grid of mid-surface points.

5.4 Validation against finite element analysis: rotating blade with prestress

[PLACEHOLDER: Table 5 — Static prestress diagnostics.] Caption: For $N \in \{1000, \dots, 8000\}$ rpm: static solve relative residual, maximum linear strain, maximum mid-surface displacement normalised by L , maximum director slope, and the resulting maximum von Mises stress at the root. Compare the root radial stress and tip displacement with the ANSYS static prestress solution (rotational inertia load). Mark the speed, if any, at which a validity threshold of Section 3.4 is exceeded.

[PLACEHOLDER: Figure 6 — Centrifugal stress field.] Caption: Contours of the mid-surface radial stress σ_{11} and shear stress σ_{12} recovered from $\mathbf{q}_0$ at 4000 rpm, showing the concentration toward the root and the chordwise redistribution induced by pre-twist and stagger. Compare with the corresponding ANSYS contour.

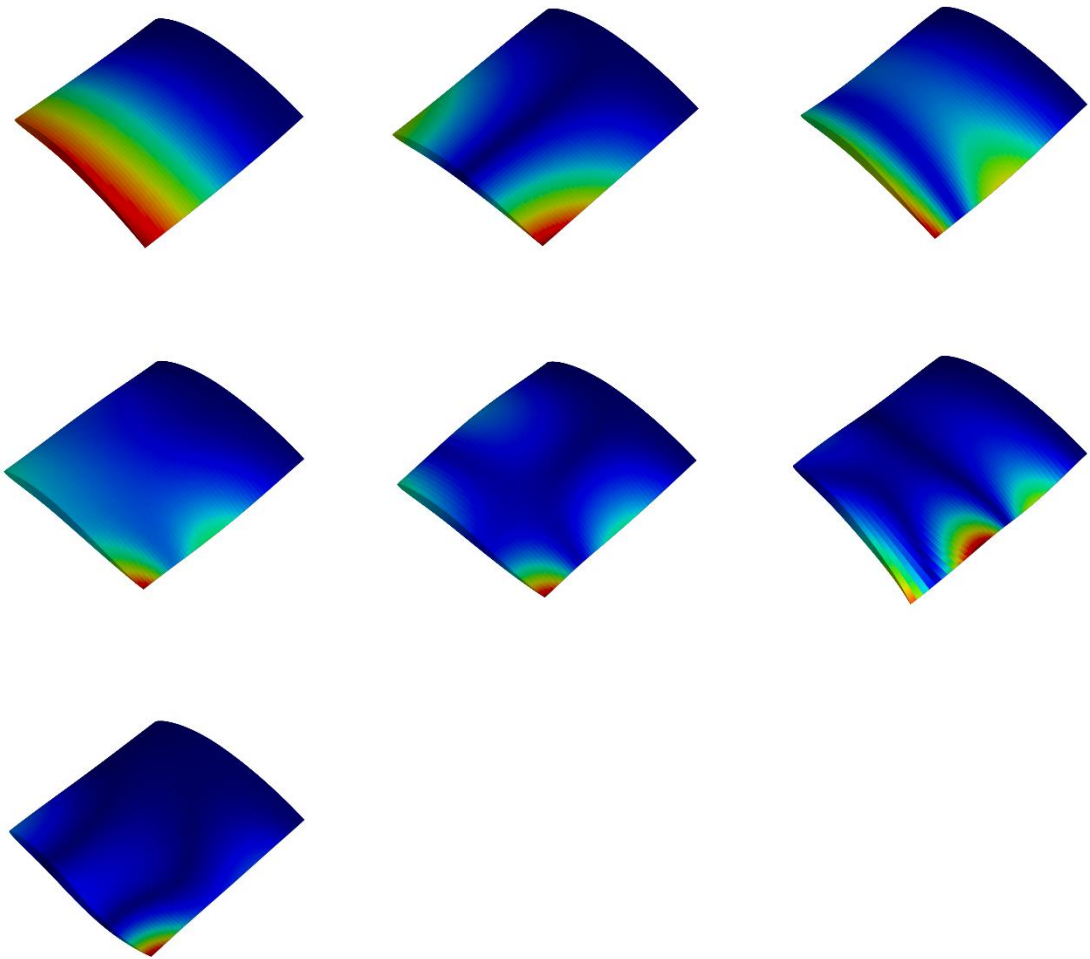
Table 6 — Rotating frequency comparison with ANSYS: Lowest seven frequencies at 4000 rpm: present method (with $\mathbf{K}_\sigma$, $\mathbf{K}_\Omega$, and $\mathbf{G}$) versus ANSYS prestressed modal analysis in the rotating frame

with Coriolis effects activated and spin softening included. Provide percentage differences and identify any modal reordering with respect to the non-rotating case.

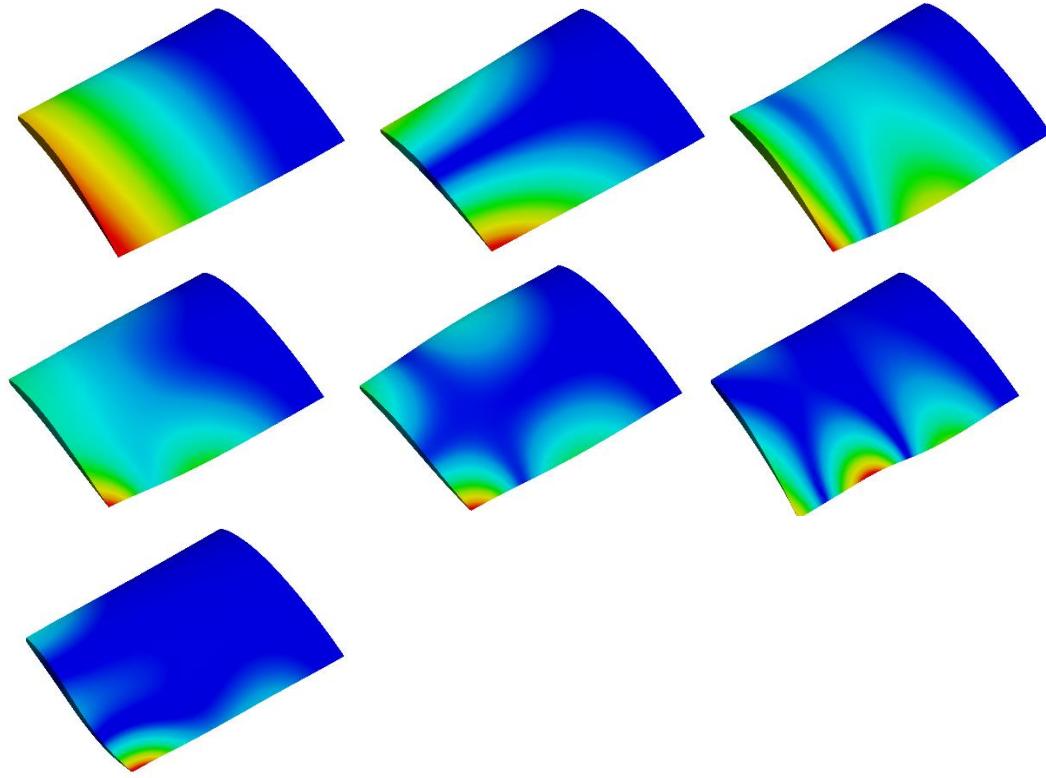
ANSYS (Hz)	Present	Error
245.85	246.012	0.07%
706.85	708.174	0.19%
970.04	969.46	-0.06%
1606.69	1606.47	-0.01%
1659.01	1657.864	-0.07%
2098.47	2116.923	0.88%
2419.68	2413.462	-0.26%

Mode Shapes

From Code:



From ANSYS:



5.5 Campbell diagram and speed-dependent behavior

[PLACEHOLDER: Figure 7 — Campbell diagram.] Caption: Natural frequencies of the lowest eight modes versus rotor speed from 0 to a maximum speed exceeding the operating range, with engine-order lines 1E to 6E. Identify the crossings of the first flap and first torsion modes with low engine orders, and mark the operating speed. Overlay the ANSYS results at discrete speeds as symbols. Discuss the different sensitivities of flap-dominated modes (strong stiffening) versus torsion- and edgewise-dominated modes (weak stiffening or softening) and relate them to the stress components entering $\mathbf{K}_\sigma$.

[PLACEHOLDER: Figure 8 — Decomposition of rotational effects.] Caption: Frequency shift of the first three modes versus speed with (a) $\mathbf{K}_\sigma$ only, (b) $\mathbf{K}_\sigma - \mathbf{K}_\Omega$, (c) $\mathbf{K}_\sigma - \mathbf{K}_\Omega$ with $\mathbf{G}$. Quantify the relative magnitudes of centrifugal stiffening, spin softening, and Coriolis coupling for a blade of this aspect ratio and stagger, and identify the speed at which the Coriolis-induced splitting of closely spaced modes becomes discernible.

[PLACEHOLDER: Figure 9 — Complex mode shapes and Coriolis coupling.] Caption: Real and imaginary parts of the mid-surface displacement for a pair of Coriolis-coupled modes at the operating speed, showing the phase lag between flap and edgewise components that distinguishes a gyroscopic (travelling) mode from a standing mode. Report the modal phase collinearity index.

5.6 Dynamic stability

[PLACEHOLDER: Figure 10 — Stability map.] Caption: Minimum eigenvalue $\lambda_{\min}(\tilde{\mathbf{K}})$ of the mass-normalised stiffened operator and maximum growth rate $\max \text{Re} \lambda$ versus rotor speed, extending to speeds

at which $\lambda_{\min}$ becomes negative and the solver switches to the general branch. Identify the divergence speed and compare it with the speed at which the ANSYS prestressed modal analysis reports a zero or negative frequency. Note that, because the present model is linear about the reference geometry, the static-validity thresholds of Section 3.4 will typically be exceeded before divergence; discuss the implications for interpreting the predicted divergence speed as an upper bound on the linear-model validity rather than as a physical limit.

[PLACEHOLDER: Table 7 — Growing modes.] Caption: At the first speed for which unstable eigenvalues are detected: list each growing eigenvalue (real part in s^{-1} , imaginary part in Hz), its residual, and the dominant character of its mode shape.

5.7 Parametric studies

[PLACEHOLDER: Figure 11 — Effect of pre-twist.] Caption: Lowest six frequencies at 0 and 4000 rpm versus total pre-twist $\theta_{\text{tip}} \in [-45^\circ, 0^\circ]$. Show the increasing flap–torsion coupling and the frequency veering between the second flap and first torsion branches.

[PLACEHOLDER: Figure 12 — Effect of taper and camber.] Caption: Frequencies versus (a) thickness taper $\tau_h \in [0, 0.7]$, (b) chord taper $\tau_c \in [0, 0.4]$, and (c) camber, comparing NACA 0012, 2412, 4412, and 6412 sections at fixed thickness. Discuss the opposing influence of camber on chordwise bending stiffness and torsional inertia.

[PLACEHOLDER: Figure 13 — Effect of hub radius and stagger.] Caption: Rotating frequencies at 4000 rpm versus hub radius $R_h/L \in [0, 2]$ and root stagger $\theta_0 \in [-60^\circ, 0^\circ]$. Relate the hub-radius sensitivity to the radial stress level and the stagger sensitivity to the redistribution of the centrifugal load between the flap and edgewise directions and to the in-plane component of $\mathbf{K}_\Omega$.

5.8 Computational cost

[PLACEHOLDER: Table 8 — Computational effort.] Caption: Wall-clock time (single BLAS thread, specified CPU), peak memory, and degree-of-freedom count for the present method at Ritz orders 4×4 , 6×6 , 8×8 , and 10×10 , against the ANSYS shell and solid models of Tables 4 and 6 (including meshing, static prestress, and modal solve). Highlight the ratio of degrees of freedom required to achieve 0.5 % agreement on the first eight rotating frequencies.

6. Conclusion

A semi-analytical formulation has been developed for the free vibration and dynamic stability of rotating compressor blades represented as pre-twisted, tapered, cambered NACA four-digit airfoil shells. The formulation combines two elements that have hitherto been confined to separate methodological traditions. From the degenerated-continuum shell concept it takes the exact three-dimensional Jacobian mapping, so that the strain field is obtained from the true Cartesian displacement gradient without recourse to the Lamé parameters and curvature approximations of classical shell theories; from the Rayleigh–Ritz tradition it takes a global, L_2 -orthonormal polynomial basis—scaled Jacobi functions in the span that enforce the clamped root exactly, and normalised Legendre functions in the chord—so that the entire blade is represented by a few hundred degrees of freedom without a finite element mesh.

The rotational physics is treated consistently in the blade-fixed frame. The centrifugal body force is integrated to a consistent load, the three-dimensional static equilibrium is solved, and the complete recovered stress tensor drives the initial-stress stiffness through the same kinematic operator and quadrature used for the dynamic operators. Coriolis coupling and spin softening enter through a skew-symmetric gyroscopic matrix and a centrifugal stiffness matrix that include rotary inertia. A singularity-

aware chordwise quadrature restores spectral integration accuracy in the presence of the square-root leading-edge behavior of the NACA thickness law and the curvature discontinuity of the camber line.

The quadratic gyroscopic eigenvalue problem is solved by an energy-scaled reformulation whose state matrix is exactly real skew-symmetric when the stiffened operator is positive definite. The Hermitian eigenproblem that results returns purely imaginary eigenvalues, so that natural frequencies are free of the spurious real parts that contaminate generalized solvers on ill-scaled companion matrices and that would otherwise be misinterpreted as damping or growth. When positive-definiteness is lost, a time-scaled general solve preserves the growth rates that identify divergence. Every static and dynamic solution is accompanied by residual and validity diagnostics, permitting the range of applicability of the linear reference-state prestress approximation to be delimited objectively.

The verification and validation program of Section 5 establishes the convergence of the Ritz expansion and quadrature, the agreement with shell- and solid-element finite element solutions at rest and under rotation, and the ability of the method to produce Campbell diagrams and stability maps at a fraction of the finite element cost. The framework extends directly to laminated composite sections by replacing the constitutive matrix, to arbitrary section families by replacing the camber and thickness functions, and to nonlinear prestress by iterating the equilibrium solve on the deformed geometry, none of which requires modification of the kinematic or discretization machinery.

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